

# Sana Gupta

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## EDUCATION

### University of California - San Diego

San Diego, CA

*Bachelor of Science in Data Science, Minor in Cognitive Science*

*Expected June 2027*

- **Relevant Coursework:** Data Visualization, Data Management, Statistical Methods, Data Structures & Algorithms, Systems for Scalable Analytics, Probabilistic Modeling and Machine Learning, Data Mining

## EXPERIENCE

### AI Data Quality Analyst

Aug 2025 – Present

*Mercor Training AI Models*

*San Diego, CA*

- Designed scalable evaluation protocols to benchmark Generative AI models, standardizing prompt engineering workflows to enable reproducible testing across 3,000+ weekly outputs
- Standardized prompt sets and evaluation metrics to enable reproducible benchmarking across models, establishing a rigorous feedback loop to optimize model performance
- Conducted data profiling and quality analysis on multimodal outputs, identifying anomalies and resolving annotation errors to strengthen dataset reliability

### Student Statistical Data Analyst

June 2025 – Sept 2025

*Department of Statistics*

*Santa Cruz, CA*

- Engineered an ETL data pipeline to normalize 200,000+ biological health records (verbal-autopsies), establishing a consistent data schema to ensure reliability for downstream backend modeling
- Automated data integration workflows using Python scripts, improving system efficiency by 30% and implementing validation testing to ensure data integrity
- Standardized processed health records to create structured datasets, directly supporting healthcare data analytics and diagnosis recommendations

### Student Data Analyst - Research Analytics

June 2024 – Sept 2024

*Department of Ecology and Evolutionary Biology*

*Santa Cruz, CA*

- Synthesized 30 heterogeneous marine-biological datasets with public records, implementing rigorous cleaning protocols to resolve inconsistencies and optimize data fidelity
- Identified trends that enhanced workflow efficiency by 15%, and communicated insights to interdisciplinary research teams to guide future study parameters

## PROJECTS

### Predictive Modeling of Infrastructure Failure | Python, Pandas, Scikit-learn, Git, HTML

Nov – Dec 2025

- Developed a Random Forest classifier (69% accuracy) and quantified algorithmic bias through stratified error analysis, isolating a statistically significant 11% performance drop in high-density urban areas
- Conducted end-to-end data processing on a longitudinal dataset (2000-2016) of 1,500+ records; performed feature engineering to build predictive models for demographic impact

### N-Gram Language Model & Tokenization Engine | Python, Pandas, NumPy, Git

Oct – Nov 2025

- Architected a language model from scratch (no external NLP libraries) in Python; implemented a regex-based tokenization pipeline to process 75,000+ tokens, focusing on code modularity
- Built an N-gram language model to demonstrate core NLU and probability concepts; assessed model performance across n-gram orders and refined preprocessing for predictive consistency

### Probabilistic Simulation & Multivariate Statistical Modeling | R, Probability Theory

Feb – Mar 2025

- Simulated 5,000 bivariate-normal samples to examine marginal, conditional, and joint distributions; visualized density surfaces and applied covariance-based transformations

## SKILLS

**Skills:** SQL, Python, R, ETL Pipelines, Data Visualization, Hypothesis & A/B Testing, Regression Analysis

**Tools:** Pandas, NumPy, Scikit-learn, Matplotlib, Git, Excel, Bash/Shell, Linux/Unix, BeautifulSoup, Requests

**Professional:** Project Management, Cross-team Collaboration, Stakeholder Communication, Documentation